INCH-POUND

MIL-P-82810(OS)
29 December 1989
SUPERSEDING
WS 22367A
9 December 1986

MILITARY SPECIFICATION

POLYTETRAFLUOROETHYLENE (PTFE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the requirements for one type of polytetrafluoroethylene, referred to herein as PTFE.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-313	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9330

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MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1457 Polytetrafluoroethylene (PTFE) Molding and Extrusion Materials

ASTM E 300 Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

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(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. **Material.**

3.1 **Composition.** The PTFE purity shall be 99.4 percent, minimum.

3.2 **Color.** The PTFE shall be opaque and the color shall range from white to gray.

3.3 **Moisture.** The PTFE moisture content shall be 0.05 percent, maximum.

3.4 **Ash.** The PTFE ash content shall be 0.1 percent, maximum.

3.5 **Average particle size (M101).** The PTFE average particle size shall be 22 $\pm$ 5 micrometers.

3.6 **Shelf life.** The PTFE shall have an unlimited shelf life when stored in the original, desiccated, sealed containers at ambient conditions not exceeding 140°F.

3.7 **Safety.**

3.7.1 **Toxic products.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.7.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

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3.8 **Workmanship.** Workmanship shall be such that the PTFE is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table I.

4.2.1 **Lot.** A lot shall consist of one or more batches (see 6.3.1) of PTFE produced by one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of PTFE that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.2.2 **Sampling.** A 500 $\pm$ 50 gram sample of PTFE shall be withdrawn from each batch to be sampled using ASTM E 300 for solids. Unless otherwise specified in the contract or order (see 6.2 and 6.5), noncompliance with any specified requirements or presence of one or more defects shall constitute cause for rejection of the lot.

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TABLE I. Quality conformance inspection.

Examinations and Tests	Requirement	Test Method
Visual examination	3.8 and Section 5	4.3.1
PTFE content	3.1	4.3.2
Color	3.2	4.3.3
Moisture	3.3	4.3.4
Ash	3.4	4.3.5
Average particle size	3.5	4.3.6

4.2.3 **Classification of characteristics.** The characteristics verified by the inspections herein are classified as Major or Minor in accordance with WR-43. Inspections that verify Major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the inspection. Inspections which are not annotated with a classification code and serial number are classified minor.

4.3 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.3.1 **Visual examination.** Visual examination shall be made to verify conformance to the requirements of 3.8 and section 5.

4.3.2 **Polytetrafluoroethylene (PTFE) content.** The PTFE content shall be determined as follows:

$$\text{Percent PTFE} = 100 - (A + B)$$

Where: A = Percent moisture, determined in 4.3.4
B = Percent ash, determined in 4.3.5

4.3.3 **Color.** Examine for color in daylight illumination using 50 ± 10 grams of sample spread over a flat surface.

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4.3.4 **Moisture.** Moisture content shall be determined in accordance with ASTM D 1457.

4.3.5 **Ash.** A sample of 10 ± 2 grams shall be weighed and placed in a previously ignited, tared, platinum crucible. The crucible and contents shall be ignited in a muffle furnace maintained at 650°C for 30 minutes. Cool for 1 hour in a desiccator and weigh. Repeat the ignition, cooling, and weighing until the change in weight between weighings does not exceed 0.2 milligram (mg). The ash content shall be calculated as follows:

$$\text{Percent ash} = \frac{A}{W} \times 100$$

Where: A = weight of ignited sample, grams
W = weight of sample, grams

CAUTION: Toxic fumes are produced by PTFE above 250°C. Vent fumes accordingly.

4.3.6 **Average particle size (M101).** The average particle size shall be determined by a volume resistivity method (Coulter Counter) using 4 percent ammonium thiocyanate in butyl alcohol as the electrolyte. Perform test in accordance with manufacturer's instructions.

4.4 **Inspection of packaging.** The PTFE containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The PTFE shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.6. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

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b. Manufacturer's name and address.

c. Net weight of contents.

d. Shelf life expiration date.

e. Storage conditions.

f. Toxicity warning (if applicable).

g. Date of manufacture.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The PTFE is intended for use in the Standard Missile Mk 104 Dual Thrust Rocket Motor ignition compound.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Inspection sampling plan, if other than as specified in 4.2.2 and 6.5.

d. Material Safety Data Sheets required (see 3.7.2 and 6.6).

e. Special marking, if other than as specified in 5.2.2.

f. Safety precautions required (see 6.7).

6.3 **Definitions.**

6.3.1 **Batch.** A batch is that amount of PTFE made as one unit in the manufacturing process.

6.4 **Possible material.** The following materials have been found to meet the requirements of this specification. This is only given for information and is not restrictive. TL-107 and TL-115A manufactured by LNP Corporation, 412 King Street, Malvern, Pennsylvania, have been found to meet the requirements of this specification.

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6.5 **Inspection sampling plan.** The sampling plan and procedures in accordance with MIL-STD-105, using an AQL of 0.65 percent for each defect, should be considered for the examination specified in 4.2.2 when this specification is applied on a contract. If contractor sampling plans are used, they should be subject to approval by the Government and provide the same protection provided by sampling in accordance with MIL-STD-105.

6.6 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 **Safety precautions.** The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DOD FAR Supplement 28.7102.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.8 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 9330-N022)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-82810	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
POLYTETRAFLUOROETHYLENE (PTFE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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